Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/26/2009
PI Name:	Turyshev, Slava G Ph.D.		
Project Title:	Laser astrometric test of relativity (LATOR) mission studies		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	91109-8001	**Congressional District:**	29
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	04/01/2004	**End Date:**	03/31/2005
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2005 (from 9/30/2007) per PI (1/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Shao, Michael (JPL) Nordtvedt, Kenneth (Northwest Analysis)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In 1915 Einstein's general relativity theory of gravity began with its empirical success of explaining the anomalous perihelion precession of Mercury's orbit, using no adjustable theoretical parameters. Shortly thereafter, the 1919 observations of star lines-of-sight during a solar eclipse confirmed the doubling of the deflection angles predicted by general relativity as compared to Newtonian arguments or the predictions from Einstein's original Equivalence Principle for local deflection in gravity. From these beginnings, general relativity has been verified at ever higher accuracy; to ~0.1% with the measurements of the Viking probe to Mars, and to ~0.03% with recent lunar laser ranging (LLR) and astrometric VLBI observations. But the continued inability to merge gravity with quantum mechanics, and recent observations in cosmology indicate that the pure tensor gravity of general relativity needs modification or augmentation. Recent work in scalar-tensor extensions of gravity which are consistent with, indeed often part of, present cosmological models motivate new searches for very small deviations of relativistic gravity in the solar system, at levels of 10^(-5) to 10^(-7) of the post-Newtonian effects. This will require a several order of magnitude improvement in experimental precision from present tests. We propose to study and here describe a new test of relativistic gravity in the solar system using laser interferometry between two micro-spacecraft whose lines of sight pass close by the Sun (LATOR). This mission will not only measures the key parameterized post-Newtonian (PPN) gamma to unprecedented levels of precision*, it also reaches ability to modestly measure the next post-Newtonian order (1/c^4) of light deflection resulting from gravity's intrinsic non-linearity. The Sun's quadrupole moment parameter, J2, will be measured with high precision, and a variety of other relativistic effects on light propagation will be measured in this experiment as well. LATOR will lead to very robust advances in the tests of fundamental physical law: this mission could discover a violation or extension of general relativity, or reveal the presence of an additional long range interaction in physical law. There are no analogs to the LATOR experiment; it is unique and is a natural culmination of solar system gravity experiments. *The Eddington parameter gamma, whose value in general relativity is unity, is perhaps the most fundamental PPN parameter, in that (1-gamma) is a measure, for example, of the fractional strength of the scalar gravity interaction in scalar-tensor theories of gravity. Within perturbation theory, all other PPN parameters to all relativistic orders collapse to their general relativistic values in proportion to (1-gamma).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Funding was for one year only. The most significant results of our LATOR mission study: There is a significant advantage in using optical wavelengths as opposed to the microwaves, the present navigational standard. This is based on the fact that solar plasma effects decrease as lambda 2 and, in the case of LATOR, we gain a factor of 10(10) reduction in the solar plasma optical path fluctuations by simply moving from lambda = 10 cm to lambda = 1 μm. Another LATOR advantage is its independence of a drag-free technology. In addition, the use of a redundant optical truss offers an excellent alternative to an ultra-precise orbit determination. This feature also makes LATOR insensitive to spacecraft buffeting from solar wind and solar radiation pressure. Furthermore, the use of existing technologies, laser components and spacecraft make this mission a low cost experiment. Thus, 1 W lasers with sufficient frequency stability and > 10 years lifetime already developed for optical telecom and also are flight qualified for SIM. Additionally, small optical apertures approaching 10–20 cm are sufficient and provide this experiment with a high signal-to-noise ratio. There also a significant advantage in using components with no motorized moving parts. This all makes LATOR an excellent candidate for the next flight experiment in fundamental physics. LATOR is envisaged as a partnership between NASA and ESA wherein both partners are essentially equal contributors, while focusing on different mission elements: NASA provides the deep space mission components and interferometer design, while building and servicing infrastructure on the ISS is an ESA contribution. The NASA focus is on mission management, system engineering, software management, integration (both of the payload and the mission), the launch vehicle for the deep space component, and operations. The European focus is on interferometer components, the initial payload integration, optical assemblies and testing of the optics in a realistic ISS environment. The proposed arrangement would provide clean interfaces between familiar mission elements. This mission may become a 21st century version of the Michelson-Morley experiment in the search for a cosmologically evolved scalar field in the solar system. As such, LATOR will lead to very robust advances in the tests of fundamental physics: it could discover a violation or extension of general relativity, and/or reveal the presence of an additional long range interaction in the physical law. With this mission testing theory to several orders of magnitude higher precision, finding a violation of general relativity or discovering a new long range interaction could be one of this era’s primary steps forward in fundamental physics. There are no analogs to the LATOR experiment; it is unique and is a natural culmination of solar system gravity experiments.
Bibliography Type:	Description: (Last Updated: 01/26/2009)
Books/Book Chapters	Turyshev SG, Shao M, Nordtvedt KL Jr. " Science, Technology and Mission Design for the Laser Astrometric Test of Relativity." in "Lasers, Clocks, and Drag-Free: Technologies for Future Exploration in Space and Tests of Gravity." Ed. H. Dittus, C. Laemmerzahl, S. Turyshev. New York : Springer Verlag, 2006. p. 429-493. http://arxiv.org/abs/gr-qc/0601035v2 , May-2006